

WORK DELIVERY REPORT

NDU Project Change Management Module — Delivery Report

Status of the work delivered for the NDU Project application. The entire Change Management module is done — the full change control loop is implemented, persisted to Firestore, and ready for approval review.

PREPARED FOR

NDU Project delivery review

PREPARED ON

September 7, 2026

CHANGE MANAGEMENT MODULE

DONE — 100%

DELIVERY BRANCH

staging

This report documents the work delivered, confirms completion of the Change Management module, and lists the code references used to verify each claim.

1. Executive Summary

CHANGE MANAGEMENT MODULE

DONE

Full change control loop delivered end-to-end

WORKSPACE TABS

8

Dashboard, Register, Impact & Approval, Audit Trail, Create, Impact Detail, Workflow, Implementation

IMPACT DIMENSIONS

15

Weighted composite impact score, cost + schedule totals

CHANGE CONTROL SURFACE

15 / 10 / 10

15 change types, 10 lifecycle statuses, 10 approval roles

Status: the entire Change Management module is **done**. Every stage of the governance loop — change request creation, 15-dimension impact assessment, multi-step approval workflow, re-baseline detection, audit trail, and implementation with rollback — is built, wired to Firestore, and protected by security rules. No part of the module is stubbed, demo-only, or pending.

The NDU Project application has been delivered across the full delivery lifecycle: initiation and business case, framework selection, planning, WBS, front-end planning, design and technical alignment, execution planning, schedule, cost estimate, procurement and contracts, risk and issue management, stakeholder engagement, SSHER and security, project controls, AI-assisted generation, program/portfolio dashboards, and the admin and subscription domain.

This report focuses on the **Change Management module** — the most recent module to reach completion — and summarizes the work delivered around it. The module runs inside the Project Controls workspace as `change-management-module`, is embedded in the Planning Phase sidebar, and is also reachable from Cost Analysis and the Project Controls hub.

2. Work Delivered — Overall Application State

The table below summarizes the delivered surface of the application. Every module listed is live on the staging branch; Change Management is called out as the module now confirmed fully complete.

Area / module	What is delivered	Status
Initiation & Business Case	Project charter, business case, solution options, cost estimate, stakeholders, opportunities, requirements	LIVE
Framework & Planning	Delivery-model selection (Waterfall / Agile / Hybrid), planning workflows, FEP suite with AI-assisted scenario planning	LIVE

	and Project Controls, Simple/Advanced views	LIVE
Design & Technical	Design management, specifications, technical alignment, architecture, deliverables and deliverable status tracking	LIVE
Execution	Execution plan, technology planning, team management, issue management, lessons learned, handover and closure	LIVE
Commercial	Procurement items, RFQs, purchase orders, contracts, vendor tracking, cost analysis with BOE and stakeholders	LIVE
Schedule	Activity schedule populated from WBS, purchase-cost badges, actual-vs-planned gap analysis, guidance assets	LIVE
Risk, Issue, Stakeholder, SSHER, Security	Risk assessment, issue management, stakeholder management, SSHER, security management with access logs	LIVE
Project Controls & KPI Dashboards	9-tab project-controls hub, baseline management, contingency/reserve tracking, program/portfolio dashboards	LIVE
Change Management	Full change control loop — 8 workspace tabs, impact assessment, approvals, re-baseline, audit, implementation	DONE
AI & Platform	KAZ AI assistant, AI-assisted generation across modules, GPT-5.x stack via secure proxy, self-hosted LLM option	LIVE
Admin & Billing	Admin domain (users, projects, coupons, subscriptions, content), survey responses wired to Cloud Functions	LIVE

Recent delivery wave (staging): migration of the AI stack to GPT-5.x models with normalized payloads (max_completion_tokens + reasoning_effort), a self-hosted LLM server option (Ollama + auth-gate), Firebase reliability and security hardening, the 22-point UX overhaul (modals, steppers,

spotlight walkthrough, yellow re-theme), dark mode support, WBS-driven epics and schedule sync controls, and persistence of cost-estimate BOE and stakeholders.

3. Change Management Module — What Was Delivered

The module lives in the Project Controls workspace and is rendered as eight tabs

(`change_management_module_screen.dart:4`). It serves both governance styles: strict change control for Waterfall delivery and routine-refinement change handling for Agile delivery.

Workspace tab	Delivered capabilities
1. Dashboard	KPI cards (pending approvals, re-baselines, average approval cycle, re-baselines this quarter), contingency/reserve tracking, 7-day CR-volume sparkline, impact summary, changes-by-category breakdown
2. Change Register	Full register with filters (status / priority / search), composite-impact badges, cost and schedule impact columns, status-colored rows, detail panel with approval workflow visualization
3. Impact & Approval Summary	Consolidated impact assessment (15 dimensions) with horizontal bar chart and per-CR cost/schedule impact
4. Audit Trail	Timeline of every action with actor/action/date-range filters and one-click CSV export
5. Create CR	Multi-section creation form: title, description, change type, priority, business justification, root cause, requested completion, emergency flag, deliverables added/modified/removed, initial cost estimate, schedule days impact, contingency/reserve drawdown, alternatives considered
6. Impact Detail	Interactive 15-dimension impact grid — each dimension scored 0–5 with schedule days, cost amount, critical flag, narrative, owner, and due date; recomputes the composite score live
7. Workflow	Approval Workflow Builder — add/remove steps, assign one of 10 approval roles per step, record per-step decisions, finalize the workflow
8. Implementation	Implementation & baseline tracker with apply-to-baseline and rollback; updates BAC, scope hash, and baseline finish; tracks contingency/reserve drawdown

What "done" means here

- All 8 tabs render with real data from Firestore — no phantom or demo seed data.
- The full change lifecycle is executable end-to-end inside the UI.
- Impact scoring, approval decisions, baseline revisions, and audit entries persist across sessions.
- Firestore rules scope change data to the authenticated user and, at project level, to requester + project owner + admins.
- The module is reachable from the sidebar, the Project Controls hub, and Cost Analysis routing.

MODULE FOOTPRINT

`change_management_module_screen.dart` — 5,460 lines (UI, 8 tabs)

`change_management_provider.dart` — 1,344 lines (state, Firestore read/write sync, metrics)

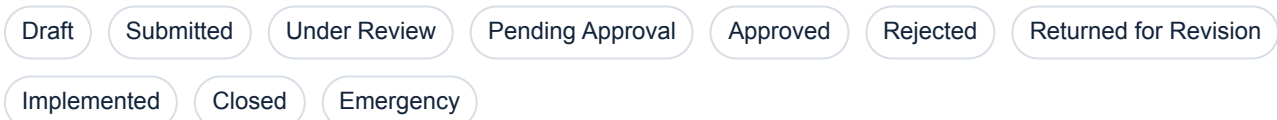
`change_management_models.dart` — data model (change types, statuses, impact, approvals, baseline history)

`new_change_request_dialog.dart` — 589 lines (Project Controls inline CR creation)

4. How the Change Control Process Works

4.1 Change request lifecycle

Each change request moves through ten explicit statuses (`change_management_models.dart:90`):



4.2 Change types

Fifteen change categories are supported (`change_management_models.dart:12`): Scope, Schedule, Cost, Resource, Procurement, Contract, Risk, Quality, Regulatory, Product, Requirements, Technical, Defect, Compliance, and Operational — each with its own icon and color treatment.

4.3 Impact assessment — 15 dimensions

Every change is assessed across 15 dimensions (`change_management_models.dart:206`): Scope, Schedule, Cost, Resources, Procurement, Contracts, Risks, Quality, Safety, Stakeholders, Funding, Benefits, Dependencies, Interfaces, and Technical. Each dimension carries an impact level (0–5 slider), optional schedule days and cost amount, a critical flag, narrative, owner, and due date.

- **Composite impact score** — weighted average across all 15 dimensions; Scope, Schedule, and Cost carry double weight (2.0) as primary constraints; result is in [0, 5].
- **Total cost impact** and **total schedule impact** — summed across approved/implemented changes for dashboard visibility.
- **Re-baseline trigger** — any change with cost or schedule impact > 0 flags *requiresRebaseline*, feeding the re-baseline detection on the dashboard.

4.4 Approval workflow

Approvals run through a step-by-step workflow builder (`change_management_models.dart:384`). Ten approval roles are available: Project Manager, Project Controls, Engineering Lead, Quality Manager, Procurement Lead, Finance, Sponsor, Safety Officer, Regulatory Authority, and Change Board. Each step records one of seven decisions: pending, approved, rejected, request info, return for revision, delegated, or escalated. The workflow can be finalized once all steps are decided, and the decision history is preserved.

4.5 Emergency changes and Agile mode

- **Emergency changes** are flagged at creation, automatically get Emergency priority, and enter the Emergency status lane so they can be fast-tracked through the workflow.
- **Dual delivery-model support** — Waterfall projects follow a strict management-of-change process; Agile projects can register routine refinements separately from controlled baseline changes.

4.6 Implementation and baseline control

- **Apply to baseline** creates a baseline revision, updates the current BAC, scope hash, and baseline finish date, and draws down contingency (\$500K default) and management reserve (\$1M default).
- **Rollback** reverts the baseline to the previous revision and restores contingency/reserve.
- Baseline history is recorded per revision with revision date, and the dashboard tracks re-baseline count per quarter plus average approval cycle time.

Why this is a complete loop: request → assess (15 dimensions) → approve (multi-step workflow) → audit (trail + CSV) → implement (apply/rollback) → re-baseline (revision history). No stage depends on a future feature; all of it is usable today.

5. Data Model & Security

Persistence

- Change requests, audit trail, and baseline history live under the user's Firestore domain (`users/{uid}/changeManagement/`) with real-time listeners.
- Provider loads on init and falls back to an empty state — it never invents phantom or demo change requests (`change_management_provider.dart:1`).
- CR numbers are generated sequentially per year (`CR-2026-001` style), and every create/update is written through to Firestore.

Security rules

- User-scoped collections (`changeRequests` , `cmAuditTrail` , and `cmBaselineHistory`) are read/write restricted to the authenticated owner (`firestore.rules:41`).
- The project-level `change_requests` collection is scoped to the requester plus the project owner and admins (`firestore.rules:263`) — the previous any-authenticated-user read was removed.

6. AI Integration

- **Auto-population from project context** — the module seeds change requests from the project's accumulated constraints and delivery context (`buildAccumulatedContext('change_management')`), giving the register a grounded starting point instead of blank rows.
- **AI recommendations** — the Change Management tab inside Project Controls surfaces AI-generated recommendations derived from project constraints.
- **Secure AI stack** — AI features run on the migrated GPT-5.x stack: normalized payloads (`max_completion_tokens` + `reasoning_effort`) via the secure `openaiProxy` Cloud Function, with an optional self-hosted LLM server (Ollama + auth-gate) for provider-outage resilience.

7. Verification & Quality

- The module ships with its state management isolated in a dedicated provider with getter-computed metrics (open CRs, pending approvals, approval cycle time, quarterly re-baselines, total cost/schedule impact), so dashboard KPIs are derived, not hard-coded.
- **Persistence is wired end-to-end** — CR creation, every update, every audit entry, and every baseline revision is written to Firestore at the mutation funnel points (`createChangeRequest`, `_updateCR`, `_addAudit`, `applyToBaseline`, `rollback`), and dashboard state (contingency, reserve, BAC) is recomputed from persisted data on reload.
- The project passes `flutter analyze` with zero errors, and the full test suite is green (164/164 tests) on the staging branch — verified at the time of this report.
- The repository contains 726 commits; 74 of them touch the Project Controls area where Change Management lives, including the world-class UI pass applied to all remaining Project Controls tabs.
- Recent commits confirm the surrounding delivery wave is green: GPT-5.x AI migration, schedule/UX refinements, cost-estimate persistence, and staging workflow expansion.

8. Approval Recommendation

Approve the Change Management module as delivered

1. Change Register with full lifecycle management (10 statuses, 15 change types, emergency lane)
2. 15-dimension impact assessment with weighted composite score and re-baseline triggers
3. Multi-step approval workflow (10 roles, 7 decision types, workflow builder)
4. Audit trail with filters and CSV export
5. Implementation tracker with apply-to-baseline and rollback, contingency/reserve drawdown
6. Firestore persistence with authenticated, owner-scoped security rules

Why approve now: the module is feature-complete, persists real data, is secured at the rules layer, and demonstrates the complete change governance loop for both Waterfall and Agile delivery

models. Follow-on work (e.g., RBAC enforcement from the earlier role recommendation report) can build on it without changing the module's shape.

9. Sources

Codebase references reviewed

- `lib/project_controls/screens/change_management_module_screen.dart:1` - module doc: 8 tabs and full feature list (dashboard KPIs, register, impact, audit, create, detail, workflow, implementation).
- `lib/project_controls/providers/change_management_provider.dart:20` - provider with Firestore-backed change requests, audit trail, and baseline history.
- `lib/project_controls/providers/change_management_provider.dart:134` - `createChangeRequest` with full payload (type, priority, justification, emergency, impact, approval steps, drawdowns) — persists the new CR to Firestore.
- `lib/project_controls/providers/change_management_provider.dart:682` - `_updateCR` funnel persists every CR update (approvals, impact edits, workflow decisions, implementation tasks).
- `lib/project_controls/providers/change_management_provider.dart:688` - `_addAudit` persists every audit entry to Firestore.
- `lib/project_controls/providers/change_management_provider.dart:330` - baseline revision creation on apply-to-baseline, persisted by version.
- `lib/project_controls/providers/change_management_provider.dart:1193` - Firestore write path for change requests (`_saveCRtoFirestore`).
- `lib/project_controls/providers/change_management_provider.dart:1303` - fire-and-forget persistence helpers (CR / audit / baseline) wired into all mutation funnels; rollback deletes the revision document.
- `lib/project_controls/models/change_management_models.dart:12` - 15 change types.
- `lib/project_controls/models/change_management_models.dart:90` - 10 change request statuses including Emergency.
- `lib/project_controls/models/change_management_models.dart:163` - 7 approval decisions (approved, rejected, request info, return for revision, delegated, escalated, pending).
- `lib/project_controls/models/change_management_models.dart:206` - 15 impact dimensions and `FullImpactAssessment`.
- `lib/project_controls/models/change_management_models.dart:378` - weighted composite impact score (Scope/Schedule/Cost at weight 2.0).
- `lib/project_controls/models/change_management_models.dart:384` - 10 approval roles for the workflow builder.
- `lib/widgets/new_change_request_dialog.dart:233` - inline CR creation dialog in Project Controls.
- `firestore.rules:41` - user-scoped change-management subcollections (`changeRequests`, `cmAuditTrail`, `cmBaselineHistory`).
- `firestore.rules:263` - project-level `change_requests` scoped to requester + owner + admins.
- `lib/routing/app_router.dart:304` - change-management and change-management-module routes.
- `lib/screens/cost_analysis_screen.dart:1825` - navigation from Cost Analysis into the module.

Repository history reviewed

- `git log --oneline -- lib/project_controls/` - 74 commits; module built through repeated delivery waves (world-class tab UI pass, collapsible headers, cross-section linkage, 22-point UX overhaul).
- `git log HEAD` - 726 total commits on the staging branch; latest: GPT-5.x AI stack migration and schedule/UX refinements (85ed63b2).